

Roaming for Applications: Federating Platforms Across Operators

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Group Size: 4 to 5

Abstract:

When you travel abroad, your phone still works, even though you are not a customer of any local operator there. Your own operator and a local one already agreed, behind the scenes, to hand you over to each other. This is roaming. This project explores what it takes to do the same thing for an application instead of a phone call. The goal is to keep a piece of software running close to a user, no matter which operator's territory that user moves into. Students will build a small system of independent, cooperating operator platforms that hand an application over to each other automatically. The mechanism is based on IPCEI-CIS, a European reference architecture that standardizes this kind of federation, and on GSMA's Operator Platform framework, which the reference architecture points to as one way of implementing it.

Context and Description

A company may want its application to run physically close to each of its users to make it feel fast and responsive. Therefore, it must avoid relying only on one distant, centralized server. Running an application close to the user like this is generally known as edge computing. If every company had to negotiate individually with every telecom operator in every country, this would be an enormous effort. A better approach is an agreement between operators, so that one operator can automatically hand over the job of running an application to a partner operator. In this scenario, the company would only have to deal with its own home operator.

Europe is tackling exactly this problem through IPCEI-CIS. Cloud and telecom providers from more than a dozen countries are collaborating under this Important Project of Common European Interest to agree on one common reference architecture for cloud-edge platforms. The goal is that a service built by one provider can interoperate with a service built by another, instead of every provider reinventing its own incompatible system. This shared blueprint is called the IPCEI-CIS Reference Architecture, or ICRA.

The ICRA defines the roles and components federated cloud-edge platforms need. One of them is the Federation Manager, the component responsible for talking to a partner platform, onboarding it, and coordinating resources and services across the two. Another is the Federation Catalog, which lets a provider publish its own offerings and discover a partner's. Together, components like these allow independent providers to cooperate without merging their infrastructures.

For mobile network operators specifically, the ICRA points to an existing industry framework called the GSMA Operator Platform. The Operator Platform defines how two independent operators can cooperate to run an application across their networks. The interface a Federation Manager uses to talk to a partner operator's platform is called the East-West Bound Interface (EWBI), and it is directly named by ICRA as one of the technologies that can realize the described kind of federation.

In this project, you will build a small, simplified version of this federation. Two or three independent services will each act as an operator platform, deployed on the cloud infrastructure provided for the project. Each platform will include a Federation Manager component. A simulated application developer will ask one of these platforms to run their application near a given location. If a platform cannot serve the request itself, its Federation Manager will contact a partner platform's Federation Manager, using the interfaced you design and implement, modelled after the real GSMA EWBI specifications. The partner platform then takes over the request. From the developer's point of view, all these activities should be transparent.

All services and components shall be made available on Instituto de Telecomunicações' cloud infrastructure, which will also be made available for development and validation activities.

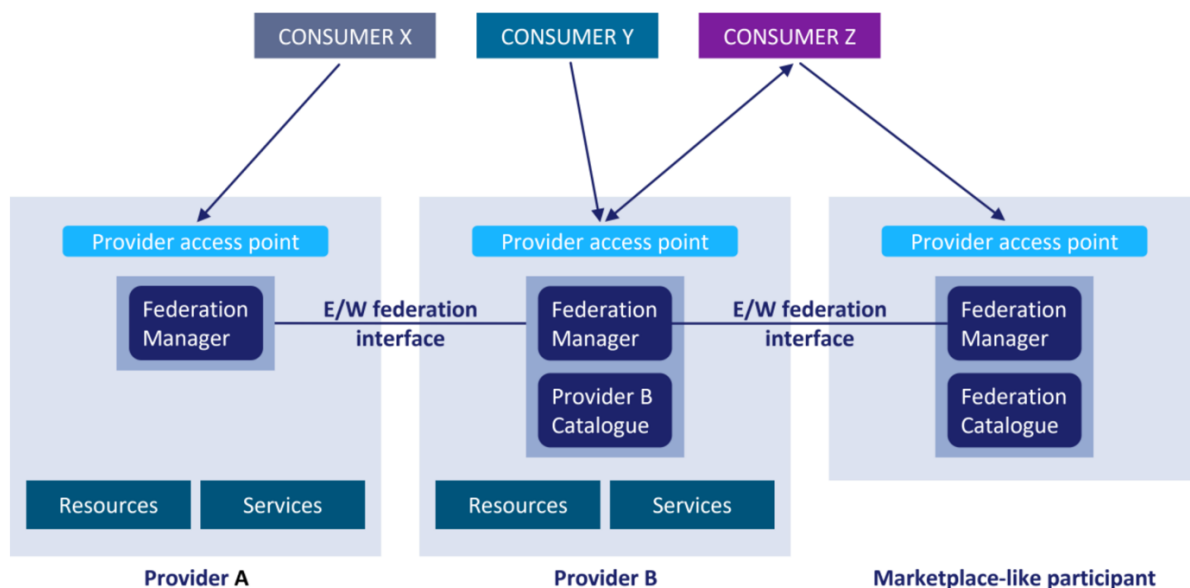


Figure 1 - Example set-up of a federation with three providers. Their resources and services are discoverable through the Federation Catalogues. Consumers acquire resources and services through the Provider Access Point. The Federation Managers communicate with each other through an East/West (E/W) interface to set up the federation and take care of ordering and provisioning.

Objectives:

- Understand how a real interoperability standard used by mobile network operators works, including the GSMA Operator Platform and its East-West Bound Interface.
- Understand the trade-offs between centralized and decentralized approaches to service discovery and hand-over between independent, cooperating systems that share no common owner.
- Design and build a system made of several independent services that must discover each other and cooperate.
- Design and Implement a Federation Manager component that decides, automatically, whether a request can be served locally or must be handed to a partner.
- Get comfortable reading and implementing against real specification documents, rather than an assignment written from scratch for a course.
- End the project with a working, end-to-end demonstration, where an application requested through one platform runs, transparently, on a different one.

Activities to Complete:

- Explore the IPCEI-CIS Reference Architecture, focusing on the Federation chapter, which describes the Federation Manager, the Federation Catalog, and the Provider Access Point.
- Explore the GSMA Operator Platform specifications, focusing on OPG.02, which defines the overall architecture, and on the EWBI-related specifications.
- Design a developer-facing API for a single platform instance, allowing an application to be registered and deployed with basic placement requirements, such as a target location and a latency constraint.
- Design and implement a Federation Manager component for each platform instance, exposing and consuming a simplified East-West Bound Interface modelled on the real GSMA EWBI.
- Implement discovery and onboarding between two platform instances. A platform must be able to register a partner and confirm that partner is reachable before attempting a hand-over.
- Implement the hand-over decision logic. A platform instance must only hand a request to a partner when it cannot serve it with its own local resources.
- Implement the hand-over itself. The partner platform should instantiate the application and report its status back through the Federation Manager.
- Package each platform instance as an independent, containerized service. Deploy the full system on the cloud infrastructure provided for the project and demonstrate the complete hand-over flow end-to-end.

While completing these activities, students will produce open-source code that can possibly be incorporated into European open-source projects under the umbrella of the European Telecommunications Standards Institute (ETSI). This gives students the opportunity to contribute to a real, active European open-source community, alongside the engineers and researchers already building these platforms.

Suggested Tech Stack:

- Python or Go
- FastAPI or an equivalent REST framework
- Kubernetes, to package and deploy every operator-platform instance directly
- OpenAPI or Swagger, to formally document the interfaces you design
- All tools used in this project must be open source (or, at least, free)

Expected Outcome at the End of the Project:

- A working system of two to three independent, Kubernetes-deployed operator-platform services, each running its own Federation Manager implementation.
- A documented, EWBI-inspired East-West interface specification describing the messages exchanged between Federation Managers.
- A live, end-to-end demonstration in which an application registered on one platform is handed over automatically and transparently to a partner platform.
- The source code and a technical report describing the work achieved.